

POULTRYWISE

Layer Farming Masterclass

A Complete Step-by-Step Guide to Commercial Egg Production

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A Professional Guide for Modern Small and Commercial Poultry Farmers

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About Book 2

This masterclass provides comprehensive operational, nutritional, and financial management guidelines specifically focused on egg-laying hens (layers) to maximize hen-day production and farm profitability.

Chapter 1: Introduction to Commercial Layer Farming

Commercial layer farming focuses exclusively on raising chickens for egg production. Unlike broilers, layers require a long-term operational commitment, with birds staying on the farm for 72 to 100 weeks.

Key Advantages:

- **Continuous Cash Flow:** Eggs provide daily income once production begins.
- **High Market Demand:** Eggs are an affordable protein source consumed globally.
- **Salvage Value:** Spent hens can be sold for meat at the end of their laying cycle.

Chapter 2: Selecting the Best Commercial Layer Breeds

Selecting the right breed determines egg yields, shell strength, feed conversion efficiency, and flock longevity.

• Isa Brown

- Brown
- 320 – 350
- Highly adaptable, calm temperament, excellent egg quality.

• Lohmann Brown

- Brown
- 310 – 340
- High feed efficiency, strong shell structure.

• Bovans Brown

- Brown
- 300 – 330
- Robust health, high production persistence.

• Hy-Line W-36

- White
- 320 – 350
- Low feed consumption, compact body size.

Breed	Primary Egg Color	Average Annual Eggs	Key Characteristics
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Chapter 3: Housing Systems & Cage vs. Deep Litter

Layer housing directly influences bird health, labor productivity, and egg cleanliness.

- **Battery Cage System:** Higher stocking density, easier egg collection, cleaner eggs, lower disease risks from manure contact, but higher initial capital cost.
- **Deep Litter System:** Lower initial setup cost, better animal welfare perception, but requires intensive litter management and higher labor for egg collection.

Chapter 4: Pullet Rearing Management (Weeks 0–17)

The rearing phase sets the foundation for lifetime egg output. A bird that does not meet target body weight before lay will underperform throughout her life.

- **Brooding (Weeks 0–6):** Focus on temperature control, starter feeds, and frame development.
- **Growing Phase (Weeks 7–15):** Uniformity is critical; monitor weight weekly to ensure 85%+ uniformity across the flock.
- **Pre-Lay Phase (Weeks 16–17):** Transition to pre-lay feed rich in calcium to build medullary bone reserves.

Chapter 5: Peak Egg Production & Lighting Management

Lighting stimulates the pituitary gland to produce hormones responsible for egg production.

- **Light Duration:** Increase light gradually up to 16 hours daily as birds reach 5% lay.
- **Never Decrease Light During Lay:** Decreasing light length during the laying phase causes a sharp drop in egg production.
- **Peak Lay:** Expect peak production (90%+ lay rate) between weeks 26 and 32.

Chapter 6: Layer Feed Formulation & Calcium Management

Feed accounts for the majority of production expenses in a layer farm.

Nutritional Stages:

- **Chick Starter:** 20% Crude Protein (Weeks 0–6).
- **Grower Mash:** 16% Crude Protein (Weeks 7–15).
- **Layer Mash:** 16–17% Crude Protein + 3.5%–4.2% Calcium (Week 18 onwards).

Calcium Tip:

Provide coarse limestone or oyster shells in the late afternoon. Eggshells form overnight, and coarse particles stay longer in the gizzard to supply calcium continuously.

Chapter 7: Disease Control & Biosecurity for Layers

Layers require continuous protection against diseases that cause drop in egg production or mortality.

- **Essential Vaccinations:** Marek's, Newcastle Disease, Gumboro (IBD), Infectious Bronchitis (IB), Fowl Pox, and Egg Drop Syndrome (EDS).
- **Deworming:** Deworm birds every 6 to 8 weeks during the laying phase.
- **Footbaths & Visitor Control:** Strictly restrict unauthorized access to laying houses.

Chapter 8: Egg Quality, Grading & Storage

Maintaining egg freshness directly affects market value and repeat customer orders.

- **Grading:** Sort eggs by weight (Small, Medium, Large, Extra Large) and inspect for cracks or blood spots.
- **Storage Temperature:** Store eggs between 12°C and 15°C with a relative humidity of 75%–80%.
- **Shelf Life:** Eggs maintain top quality for up to 21 days when stored properly.

Chapter 9: Financial Management & Molting Decisions

Managing production costs ensures steady farm profits.

Hen-Day Egg Production (HDEP):

$$HDEP (\%) = (Total\ Eggs\ Collected\ in\ a\ Day / Total\ Live\ Birds) \times 100$$

- **Culling Strategy:** Systematically cull non-laying birds (pale combs, tight pelvic bones) to avoid wasting feed.
- **Depopulation vs. Molting:** Decide whether to sell spent hens at week 72–80 or induce molting to extend laying into a second cycle.

Chapter 10: Egg Marketing & Distribution Networks

A structured sales strategy guarantees consistent off-take.

- **B2B Contracts:** Supply bakeries, hotels, supermarkets, and catering companies.
- **Retail Packaging:** Use branded egg crates/cartons with **PoultryWise** labels to build brand recognition and command premium pricing.
- **Egg Grading Standards:** Consistently separate cracked or dirty eggs to maintain quality trust with buyers.

--- End of Book 2 (PoultryWise Layer Farming Masterclass) ---